3D Food Segmentation

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1 Introduction

By providing our bodies with the necessary nutrients and caloric intake, our diets allow us to function, shaping physical health, energy levels, and long-term disease risk. Heart disease is the leading cause of death in the United States, and it is driven largely by diet-related conditions such as obesity and type 2 diabetes. Furthermore, as diet also significantly impacts cancer and mood disorders, which affect lifespan and quality of life, dietary choices represent a powerful tool to intervene in the largest health crises. Tracking caloric and nutritional intake typically requires the use of a digital scale to weigh foods and multiply by precalculated per-gram values. However, this is an inconvenient and time-consuming process. Recent advances in computer vision have produced models that can identify food; however, estimating portions and energy content from a single photo is highly unreliable, since photos can vary widely in lighting, camera distance, angle, and intrinsics. Given that many modern handheld devices now have LiDAR depth sensors, the aim of this project is to combine an RGB camera image with a 3D scan of the food to more accurately calculate portion sizes and nutritional information. The system reconstructs a metric-scale 3D model of a plated meal, segments the individual food items, estimates each item's volume, and converts volume to calories and macronutrients. All the user has to do is briefly scan their plate with their phone. This design follows from recent evidence that metric depth is the strongest signal for accurate portion estimation.

2 Related Work

Depth-sensor calorie estimation. SNAQ [3] used the depth sensor on an iPhone, segmenting food with a CNN, fitting the plate surface with RANSAC, and uses a nutrition database as reference to calculate nutrition for segmented foods, reaching a 12.7% mean energy error across 48 hospital-kitchen meals. This idea is extended in LiDARCalorieCam [2], which estimates calories in real time without reference objects. MFP3D [5] reconstructs a point cloud from a single photo. Its ablations motivate the design of this project, where a point cloud from a full 3D scan had a 17% MAPE, compared to a 62–98 % from reconstructed point clouds along.

Vision-language models. CLIP [6] embeds images and text in a shared space, such that models can recognize categories named only in text. Large multimodal models can also estimate calories directly from a food photo, but they often misjudge volume, which drives calorie error [9].

Segmentation models. SAM 3D [7] predicts an object's full 3D shape from a single image, though surfaces the camera cannot see are generated rather than measured. Search3D [8] builds a

searchable 3D representation of a scan, such that a text query can be granular looking at objects as a whole, parts, or regions based on attributes and composition. FoodSAM [4] labels SAM's masks using the coarse output of a conventional food segmentation model, but categories are limited to a training set.

Datasets. Nutrition5k [10] contains roughly 5,000 cafeteria dishes captured by an overhead RGB-D camera, labeled with per-ingredient weights and nutrition, and shows that depth improves mass and calorie prediction. MetaFood3D [1] prov ides 3D food models with nutrition labels.

3 Problem Description

Input. A short phone scan of a meal. Each frame has a color image, a LiDAR depth map, and the camera's position, which together let the frames be combined into one 3D model of the meal.

Output. For each food item on the plate: its name, its mass in grams, and its calories, protein, carbohydrates, and fat. The target is energy error at or below the roughly 13% reported by prior depth-based systems [3].

Assumptions. Food is on a plate or other dish, and food items are relativley seperated and visible. Volume is converted to mass with standard densities, such that this variation is an accepted source of error.

Research questions.

1. How much does a handheld 3D scan reduce mass and calorie error compared with a single RGB image or a single RGB-D frame?
2. Do open-vocabulary segmentation models, developed for room-scale scenes, separate food items accurately at plate scale?
3. Does filling in hidden surfaces with a generative model such as SAM 3D improve volume accuracy over simply closing the shape at the plate?

4 Planned Approach

Capture. The user walks the phone around the plate. An app records color images, LiDAR depth, and camera positions, and fuses them into a single 3D model of the meal. One known limitation is that the LiDAR depth map is coarse: 256×192, against a high definition color image. Prior work found that this low point density causes volume overestimation for small or touching items [2]. Fusing depth across the frames of the scan should reduce this.

Segmentation. A VLM first names the foods on the plate to gather a list of foods to segment. There are two possible approaches with the first using a model like SAM 3 [11], which segments and tracks every matching item across the frames of the scan, where each mask is projected back onto the 3D model to label points. It is also possible to adapt open-vocabulary point cloud method such as Search3D [8], using each name as a text query. This uses shape as well as appearance, but these methods were developed for room-scale scenes. Comparing the two answers research question 2.

Volume estimation. The plate surface is found, and the volume of each food is measured as the space between its labeled surface and the plate. The scan covers the sides and back of each item, but not its underside or inside. SAM 3D [7], which predicts an object's full 3D shape from one image and its outline, fills in these hidden parts, scaled to match the scanned points. As a comparison, SAM 3D is also run on a single frame with no scan, to measure how much the scan adds. Accuracy is measured as volume and mass MAPE and calorie MAE, first on objects of known size, such as boxes and cans, then on weighed portions of food.

Nutrition. Each food's name is looked up in the USDA FoodData Central database for its nutrients per gram. Volume is converted to mass using a standard density for that food, and mass times nutrients per gram gives the calories, protein, carbohydrates, and fat. The totals for the meal are the sum over all items.

5 Timeline

Dates	**Milestone**
Sep 14–19	Submit proposal (Sep 17). Set up compute cluster.
Sep 20–Oct 3	Build capture app and 3D reconstruction.
Oct 4–17	Set up VLM for naming foods. Research and test segmentation approaches. Start collecting weighed meal scans.
Oct 18–31	Progress report (Oct 20). Volume estimation, checked on objects of known size.
Nov 1–14	Nutrition lookup and first end-to-end results. Continue data collection.
Nov 15–28	Evaluation and comparisons.
Nov 29–Dec 12	Final report to advisor (Dec 7) and poster (Dec 8).
Dec 13–17	Revise with advisor feedback. Submit final deliverables (Dec 17).

Table 1: Planned milestones.

6 Deliverables

- Code for the capture app and processing pipeline
- Dataset of weighed meal scans
- Demo video
- Poster
- Final report

References

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